

A Three matched intervention arms

Keyed

Delta psi changes the gate
alpha yaw high before clearance

Circular honest

Delta psi is gauge
alpha yaw remains zero

Circular placebo

15 deg yaw sweep present
but fixed across Delta psi

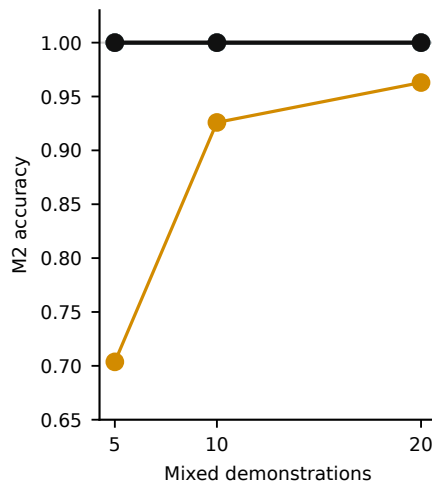
same contexts, seeds, controller, anchor and retry budget

● Frame weighted
 ● TP-GMM SE(2)
 ● Generic RBF
 ● Full operator
 ● Pdiag finite

B Generator-law error at N=20

	Keyed	Circular honest	Circular placebo
Frame weighted	0.338	0.262	0.284
TP-GMM SE(2)	0.142	0.001	0.000
Generic RBF	0.159	0.000	0.001
Full operator	0.159	0.000	0.001
Pdiag finite	0.180	0.002	0.001

C Keyed-vs-circular discrimination



D Placebo symmetry gap

